applications without the need to build and maintain their own blockchain infrastructure. It provides a platform for users to create and manage blockchain based solutions, such as smart contracts, decentralised applications (dApps), and digital currencies, using pre-built blockchain components.

BaaS providers offer a range of blockchain services, such as network deployment, smart contract management, node management, data storage, and identity management. These services are typically provided through a user-friendly interface that simplifies the process of building and deploying blockchain solutions.

BaaS is particularly useful for businesses that need to build and deploy blockchain solutions quickly and efficiently, as it eliminates the need for complex development processes and reduces time-to-market. It is also beneficial for businesses that want to ensure the security and privacy of their data, as blockchain technology provides a secure and tamper-proof way of storing and transferring data.

You can see in Figure 2 that the BaaS platform is provided by a cloud provider such as Amazon Web Services (AWS), Microsoft Azure, IBM, or any other. The cloud provider manages the blockchain infrastructure, which consists of nodes and other components needed to run the blockchain network. The BaaS platform sits on top of the blockchain infrastructure and provides an easy-to-use interface for developers to create and manage blockchain networks. It abstracts away much of the complexity of setting up and managing a blockchain network, allowing developers to focus on building applications on top of the blockchain. Finally, the blockchain network is the actual distributed ledger that runs on top of the blockchain infrastructure. This network can be configured to support various consensus mechanisms, smart contracts, and other features depending on the needs of the application being built on top of it.

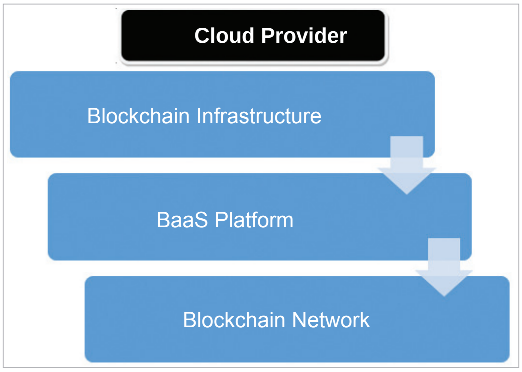

Figure 2: Layered approach with Blockchain as a Service (BaaS)

One of the key benefits of BaaS is that it allows businesses to experiment with blockchain technology without having to invest in expensive hardware or software infrastructure. It also enables them to focus on their core competencies while leaving the management of the blockchain infrastructure to a third-party provider.

Popular cloud platforms for blockchain programming

Amazon Web Services (AWS): AWS is a cloud platform that provides a wide range of services for building and deploying blockchain applications. It provides several blockchain-specific services, such as Amazon Managed Blockchain and Amazon Quantum Ledger Database, as well as more general services, such as compute, storage, and networking.

Microsoft Azure: Microsoft Azure provides several blockchain-specific services, such as Azure Blockchain Service, as well as more general services, such as compute, storage, and networking.

IBM Cloud: IBM Cloud is a cloud platform that provides several blockchain-specific services, such as IBM Blockchain Platform, as well as more general services, such as compute, storage, and networking. IBM's BaaS platform supports multiple blockchain frameworks, including Hyperledger Fabric, Ethereum, and Stellar.

Oracle Blockchain Cloud Service: Oracle provides a BaaS platform that supports Hyperledger Fabric, Ethereum, and Corda.

R3 Corda: Corda is a distributed ledger platform designed for businesses that require high-throughput, privacy, and interoperability.

ConsenSys Quorum: Quorum is an enterprise-focused version of Ethereum that is designed to support private, permissioned networks.

NEM NIS1: NEM is a blockchain platform designed for enterprise use cases, with features such as multi-signature accounts, encrypted messaging, and a reputation system.

Chain: Chain provides a BaaS platform that supports Hyperledger Fabric and Corda, with a focus on data privacy and security.

Stratis: Stratis is a blockchain platform that provides support for both private and public networks, with a focus on enterprise use cases.

Baidu Blockchain Engine: Baidu's BaaS platform supports multiple blockchain frameworks, including Ethereum and Hyperledger Fabric, and is designed to provide high scalability and reliability.

Google Cloud Platform (GCP): GCP is a cloud platform that provides several blockchain-specific services, such as Google Cloud Blockchain Templates and Google Cloud Spanner, as well as more general services, such as compute, storage, and networking.

A few other important cloud platforms for Blockchain as a Service (BaaS) are listed in Table 1.

When choosing a cloud platform for blockchain programming, it is important to consider factors such as ease of use, scalability, security, and cost. It is also important to consider the specific requirements of the project and the level of expertise of the development team.

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